

INST 327

Section: 0202

Final Report

12/10/2025

Team 2

Team lead - Emilio Sanchez San Martin, liaison - Hein Htet, backup liaison - Matthew Daniel, Andy Gunawan, Simran Shergill

Introduction:

Over the course of the semester, our team developed a relational database designed to organize and analyze crime data reported in Montgomery County, Maryland from July 2016 to January 2023. We began this project with the goal of transforming a large and unstructured dataset into a system that could support meaningful analysis for individuals, community organizations, researchers, and policymakers. Montgomery County is the most populated county in Maryland, and with over 300,000 recorded incidents during this period, understanding crime trends across such a large region requires a structured and reliable data system.

At the start of the semester, we identified several guiding research questions that shaped the design of our database. These included determining which areas of the county experience the highest crime rates, which police stations respond to specific incidents, and what types of crimes occur most frequently when filtered by offense code, district, or ZIP code. We also sought to explore whether certain crime types correlate with patrol deployment patterns, whether geographic “hotspots” can be identified for targeted prevention efforts, and how crime frequency varies across neighborhoods with different socioeconomic characteristics. These questions helped guide our choice of entities, attributes, relationships, and the queries and views supported by our final implementation.

Using data from the Montgomery County Police Department (MCPD) through the EJustice system, we cleaned and standardized the raw dataset by addressing missing values, inconsistent formatting, duplicates, and shifting offense descriptions. From this cleaned version, we designed a normalized relational schema that captures the major components of crime reporting: incidents, offenses, locations, policing districts, agencies, and associated places. This structure allows users to explore crime patterns more clearly and reliably.

Database Description:

Logical Design:

The Montgomery County Crime Database is designed to provide a comprehensive record of reported crime incidents from July 2016 to January 2023. This database consists of seven core tables: *Offense*, *Incident*, *IncidentOffense*, *Agency*, *District*, *Location*, and *Place*. The *Offense* table captures the specifics of each crime, including offense codes, NIBRS codes, crime name, crime category, and the type of victim affected. The *Incident* table represents individual crime events, linking to *Location*, *Agency*, *Place*, and the junction table *IncidentOffense*, which facilitates the many-to-many relationship between incidents and offenses, while *District* records administrative areas, including sector, beats, and PRA codes. Each *Location* is associated with a district and includes detailed address information and geographic coordinates. The *Place* table describes the type of location where an incident occurred, such as school, residence, or commercial property.

Physical Design:

When it comes to the physical design of our Montgomery County Crime Database we generated all our tables and queries in MySQL 8.0 using the built-in InnoDB storage engine, which supports transactional integrity, foreign key enforcement, and indexing. Our *inst327project* database that contains all the Montgomery County crime data is composed of seven normalized tables: *Agency*, *District*, *Location*, *Place*, *Offense*, *Incident*, and *IncidentOffense*, each defined with specific datatypes, constraints, and indexes. All core entities, such as the various IDs like *offense_id*, *incident_id*, *agency_id*, *district_id*, *place_id*, and *location_id*, use unsigned integer surrogate keys with AUTO_INCREMENT features as primary keys. But with the *IncidentOffense* table, it utilizes a composite primary key to enforce and uphold the many-to-many relationship between the incidents table and offense table. We also made sure to uphold referential integrity through our foreign key constraints linking the *Incident* to *Location*, *Agency*, and *Place* and *Location* to *District*. In our database the character data is stored using UTF-8 encoding with VARCHAR fields sized such as NIBRS_code VARCHAR(20), cr_number VARCHAR(45), block_address VARCHAR(100), etc, meeting our database's attribute requirements, while our geographic coordinates dataset are stored using DECIMAL data types for precision and location accuracy. Other datatypes that we incorporated was, the DATETIME datatype for our *Start_Date_Time* and *End_Date_Time* columns to validate

dates, store universal time, and perform fast and efficient indexing and queries when filtering out or comparing dates using specific functions or running datetime calculations or complex logic. Finally we also implemented unique constraints to the critical identifiers, such as offense codes and case report numbers, to stop duplication, and made sure that the indexes that we created on foreign keys enhance and optimize our query performance.

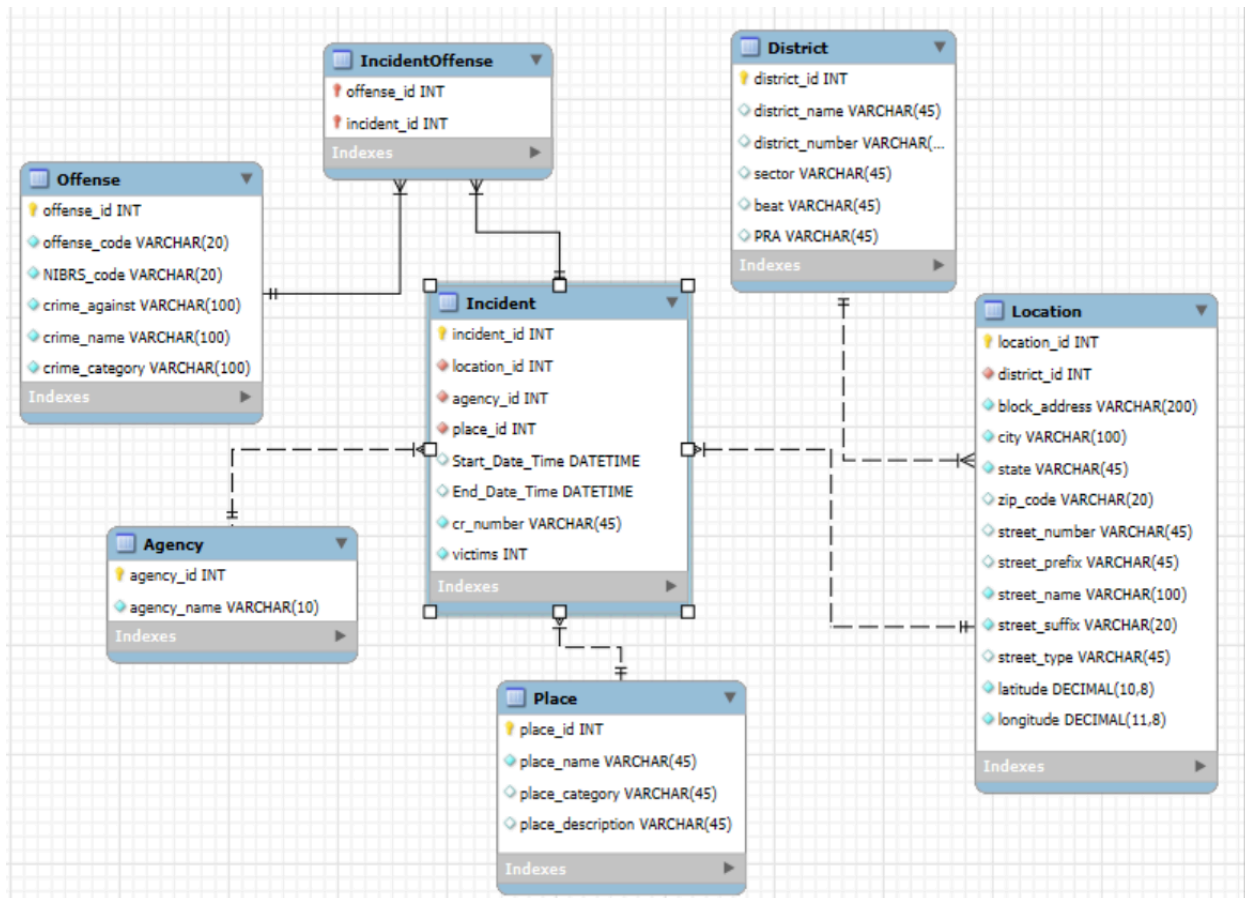


Image 1. Final ERD

Changes from Original Design:

The Raw dataset provided by the MCPD contained a large amount of unstandardized and inconsistent information, so one of our first steps was to learn and implement the steps of normalization to improve accuracy and usability. Instead of storing offense information repeatedly in every incident, we created a many to many relationship using the IncidentOffence junction table. This allowed us to more accurately represent incidents with multiple offenses and

enabled more flexible queries, such identifying all incidents associated with a particular crime category by their specific IDs.

Data cleaning also played a major role in refining the final design. There were inconsistencies in address formatting, data fields, and capitalization across attributes. During preprocessing, we standardized these elements by unifying elements like street names, city names, and other textual fields, and by reviewing the first 150 rows in detail to identify structural issues before scaling up. These cleaning steps helped reduce errors, improve data reliability, and ensure that records could be inserted smoothly into our normalized table structure.

Sample Data:

We designed a database that can be useful for any researcher or policymaker to capture crime data to obtain more insight as to the trends of crimes there are by location, time, and type of crime. Our group's goal for this project is to make it easier and understandable to gain this information from our database. There will be a mix of many different tables given places, incidents, locations, etc, to help users be able to explore insights and gain better information through this huge dataset.

Some of the data we use will fit into our database to align with our goals for this project. Since there are over 300,000 records in our raw dataset, we realized that we had to refine our database to store a smaller amount of records related to our crime data. With lots of overlap and cleaning, we created many separate tables for different attributes, such as, CrimeAgainst, Crime Name, Locations, Victims, Time, Agency, and much more to explore. This helped keep our tables clean and not keep a huge, inconsistent database to make relations and understand the information more clearly. We also cleaned the different formats or attributes since dates and time go together, geographical data with coordinates come into play, and more occurrences in crime come in one report. We all try to approach this project in a way to make it easier for users to understand patterns and look at the frequency of specific crimes in different areas or for long periods, helping us understand this complex data set and make it more meaningful to use.

Sample Query # 1

What is the total PRA in Montgomery County, and what is the average PRA for each city?

```
/*View 1 */  
CREATE VIEW v_city_pra_totals_and_averages AS  
SELECT  
    IF(district_name IS NULL,  
        "Total PRA in Montgomery County",  
        district_name)  
    AS district_name,  
    sum(PRA) AS "Total PRA in each city",  
    IF(GROUPING(district_name) = 1,  
        NULL,  
        ROUND(AVG(PRA), 1)  
    ) AS "Average PRA in Montgomery County"  
FROM district  
GROUP BY district_name WITH ROLLUP  
ORDER BY  
    IF(district_name IS NULL, 1, 0),  
    district_name;
```

Result Output:

district_name	Total PRA in each city	Average PRA in Montgomery County
BETHESDA	1556	141.5
CITY OF TAKOMA PARK	1611	805.5
GERMANTOWN	6498	541.5
MONTGOMERY VILLAGE	9284	488.6
ROCKVILLE	4357	311.2
SILVER SPRING	3839	182.8
WHEATON	6506	382.7
Total PRA in Montgomery County	33651	NULL

249 13:43:00 SELECT * FROM v_city_pra_totals_and_averages LIMIT 0, 1000 8 row(s) returned

Sample Query # 2

Which crime category had the most total victims and incidents?

```
/*View 5 */

CREATE VIEW v_victims_by_crime_category AS
SELECT
    o.crime_category AS Crime_Category,
    SUM(i.victims) AS Total_Victims,
    COUNT(i.incident_id) AS Total_Incidents
FROM Incident i
JOIN IncidentOffense io USING (incident_id)
JOIN Offense o USING (offense_id)
WHERE i.victims IS NOT NULL
GROUP BY o.crime_category
ORDER BY Total_Victims DESC;
```

Result Output:

Result Grid   Filter Rows: Search Export: 			
	Crime_Category	Total_Victi...	Total_Incide...
	DRIVING UNDER THE INFLUENCE LIQUOR	15	15
	ASSAULT - 2ND DEGREE	11	9
	LARCENY - SHOPLIFTING	8	8
	POLICE INFORMATION	8	8
	DRUGS - MARIJUANA - POSSESS	7	7
	MENTAL ILLNESS	7	7
	LIQUOR (DESCRIBE OFFENSE)	6	6
	PUBLIC PEACE - DISORDERLY CONDUCT	6	6
	LARCENY - FROM AUTO	6	6

Query Requirements Mapping

Views	Req. A	Req. B	Req. C	Req. D	Req. E
v_city_pra_totals_and_averages			X		
v_zip_alcohol_crime_breakdown	X	X	X	X	
v_high_crime_addresses	X	X	X	X	X
v_cases_by_agency_and_city	X	X	X		
v_victims_by_crime_category	X	X	X	X	

Database Ethics Considerations:

The Montgomery County Crime Database raises important ethical considerations, primarily related to data privacy, fair use, and responsible handling of sensitive information. The dataset contains detailed attributes such as zip codes, cities, addresses, and specific locations of crimes. While this information is essential for analyzing crime patterns, it also poses potential risks to individuals, businesses, and communities. For example, repeatedly labeling a particular location as a crime hotspot could harm a business's reputation or a homeowner's sense of safety. Public disclosure of precise addresses could compromise residential security or inadvertently allow malicious actors to exploit the information. Additionally, reporting on crime locations may increase anxiety among local residents, particularly if they perceive a higher level of risk in their neighborhood.

Another ethical consideration involves policing and surveillance. The dataset includes attributes such as police response area (PRA), sector, and beat, which detail police presence in specific locations. While this information is valuable for monitoring law enforcement activity and understanding patterns of crime response, it could be misused. Criminals might avoid heavily monitored areas, and excessive focus on high-crime zones could lead to overpolicing, potentially infringing on civil liberties or creating unsafe situations. At the same time, accurately monitoring crime rates and response efficiency requires precise location data, which is necessary for both criminal investigations and informed policy decisions.

Privacy and responsible use are also central considerations. Although the data is publicly available through the Montgomery County Police Department's EJustice system, location information could still expose individuals or businesses if not handled carefully. The database avoids personally identifiable information and does not attempt to link incidents to specific demographic data. Additionally, while detailed police presence data can help understand crime patterns, it could potentially be misused by criminals to evade law enforcement or by authorities to over-police certain communities. Documenting dataset limitations and emphasizing responsible interpretation helps mitigate these risks.

Finally, throughout the process of populating our Montgomery County Crime database with our cleaned up data we almost violated certain data privacy, fair use, and ethical considerations in our project's overall design when running our queries. In our initial project proposal we created a list of possible questions that our database could answer and one of the questions we asked was "What are the zipcodes in Montgomery County that are considered high middle or low income areas and display them in separate columns along with the number of crimes committed in that area?" Our database contained a list of Montgomery County zipcodes, but not their income level so we decided to conduct some research and tried to determine which corresponding zipcodes in our column were located in low, medium, or high income areas. We then established a separate column that contained the income levels when running our query. We thought this was an excellent query that answered the question and established unique data attribute relationships, but realized that we were manipulating with the data as it wasn't in our raw Montgomery County Crime dataset and there was a reason for that. Running a query that establishes a relationship between the number of crimes committed and the income level of a certain zipcode area would violate people's right to privacy and cause a lot of bias and misrepresentation targeting marginalized communities or misidentifying crimes or individuals as criminals and damaging their reputation. An individual's income is their own private financial information which must be secured and protected, which is why we had to refine the question we asked for our database to fulfill our database requirements.

Lessons Learned:

Throughout this project, our team encountered several challenges that ultimately taught us valuable lessons about database design, teamwork, and working with real-world data. One of the first obstacles we faced was the data-cleaning process. At the beginning, we tried to clean the entire dataset at once, which quickly became overwhelming. The raw dataset contained numerous inconsistencies, missing values, and unclear formatting, and attempting to fix all these issues across thousands of rows caused confusion and made progress slow. At that point, we had two options: continue struggling with the full dataset or temporarily reduce the scope to focus on a smaller subset. After consulting with our TA, we chose the second option and shifted to cleaning and testing only the first 150 rows. This decision not only made the process more

manageable but also taught us the importance of breaking large technical tasks into smaller, testable stages rather than trying to solve everything at once.

Potential Future work:

Although our current implementation successfully organizes, standardizes, and normalizes crime data from Montgomery County, there are several opportunities to extend the database to support more advanced analysis, improved usability, and broader real-world applications. Future development could include:

1. **Integrating Additional Data Sources**

One significant area for future enhancement is the incorporation of external contextual data. Examples of data that can be integrated include census demographic information, socioeconomic indicators, school zone boundaries, population density, transportation infrastructure and other data sources that will allow for more nuanced analysis of crime patterns.

2. **Implementing GIS and Spatial Analysis Features**

While our current schema includes coordinates for each location, spatial analysis is not yet fully leveraged. Future work could involve integrating the database with geographic information systems (GIS) tools and enabling features such as heat maps, distance-based queries, nearest-neighbor analysis, and cluster detection. These capabilities would make the database more valuable for law enforcement resource planning, urban studies research, and community organizations interested in understanding geographic crime trends.

3. **Enhancing Data Cleaning, Validation, and Automation Pipelines**

The preprocessing and cleaning steps completed this semester were done manually and on a limited subset before scaling. A next step would be to develop automated data-cleaning scripts or ETL (Extract, Transform, Load) pipelines that can process new crime reports in real time to improve efficiency. Future improvements could also include validation constraints, anomaly detection for outlier incidents, and automated error reporting to ensure consistent data integrity and quality as new data is added.

4. **Supporting Predictive Analytics and Machine Learning Models**

The structure of the database lends itself well to more advanced forms of analysis, including machine learning. Future work could involve exporting the cleaned, normalized data into models designed to forecast crime hotspots, predict seasonal fluctuations, or identify variables most associated with high-risk areas.

5. **Enhancing Security, Ethical Safeguards, and Controlled Access**

As the dataset includes geographically sensitive information, future improvements could introduce improved security features such as tiered access levels, anonymization layers, stricter auditing mechanisms for query usage, expanding ethical safeguards such as restricting views that overly expose specific residential addresses to ensure responsible data use and reduce potential harms associated with misinterpretation or misuse of the information present in the database.

6. **Developing a Full Application Layer for Public Use**

Finally, the database could be extended into a complete web-based publicly available crime information system. This system could allow users to run pre-built queries, visualize trends, download filtered datasets, or contribute additional context to incidents. Such an application would transform the database from a course project into a functional public resource with a positive societal impact.

References

Montgomery County Government. (n.d.). *Crime: Open data from the Montgomery County Police Department*.